PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	POSTNATAL EXERCISE INTERVENTIONS: A SYSTEMATIC REVIEW OF ADHERENCE AND EFFECT
AUTHORS	Mullins, Edward; Sharma, Shalini; McGregor, Alison

VERSION 1 – REVIEW

REVIEWER	Tan, Bee University of Leicester
REVIEW RETURNED	27-Sep-2020

GENERAL COMMENTS	This systematic review addresses a very important issue. The review is well performed, employing appropriate methodology.

REVIEWER	Kołomańska-Bogucka, Daria Uniwersytet Jagielloński w Krakowie, Department of Biomechanics and Kinesiology
REVIEW RETURNED	14-Oct-2020

GENERAL COMMENTS	The Authors present a systematic review of 24 studies about physical activity of women during postpartum period. This is a very important topic, because properly selected physical activity has a positive effect on the course of pregnancy, delivery and postpartum period. The authors have pre-registered the review on PROSPERO. They also presented a detailed way of browsing individual databases. Information on the individual phases of the study is presented in the PRISMA diagram. The studies qualified for the review are presented in detail in the tables. My comments are: 1) Page 4, verse 3-5 - the same item from the bibliography is used in the following sentences - I suggest you only include it at the end of the last sentence in this paragraph. 2) On page 4, verses 17-20, the same publication is also marked in two consecutive sentences. 3) Line 15, page 4 - "Sixty percent of women ..." where other percentages are shown, they are written in numbers 4) You refer to the fact that women's physical activity decreases each trimester "Fewer than 13% of women meet recommended PA levels during pregnancy, and this decreases with each trimester". Are You able to present these changes? 5) In the paragraph Search strategy and selection criteria you wrote that "…publication date was not restricted", but could you add information in what period (month, year) the review was performed?

REVIEWER	Swindle, Taren

	University of Arkansas System
REVIEW RETURNED	01-Dec-2020

GENERAL COMMENTS	Thank you for the opportunity to review "Postnatal Exercise Interventions: A Systematic Review of Adherence and Effect." This is a systematic review focusing on 24 RCTs of physical activity interventions during the first-year post-partum. The primary outcomes are adherence (operationalized as compliance and retention rates) and effectiveness. This systematic review showed inconsistent results for most outcomes, besides adverse events and negative impacts on breastfeeding or infant growth. There was a lack of reporting on adherence measures and conclusions are generally based on low quality studies at a high risk of bias. Appropriate detail is given for the methods of the review, and justification for exploration of this topic is clear and fitting. I have the following comments, questions, and suggestions to improve the paper: • Prospero registration was first submitted on 10/25/2018. At the time of submission, the search had been completed and other parts had been started (i.e., data extraction and risk of bias assessment). I recommend clarifying with the authors what year the search was conducted and if >12 months ago, request an updated search. • No meta-analysis was performed, and this systematic review focuses on intervention studies, yet they used the Meta-Analysis of Observational Studies in Epidemiology (MOOSE) guidelines to report findings. I believe PRISMA would be more appropriate. I recommend requesting a revised manuscript that uses PRISMA instead of MOOSE guidelines. • The authors state that sub-group analyses were conducted. It is not clear what is meant by analyses in this case. I wonder if instead it was an examination of patterns. • The authors state that "supervised studies appeared less effective than unsupervised studies." For this comparison and others in the studies, could the authors be clear about the thresholds or specific data used to draw these conclusions? • The take home messages do not include any findings from the study. • The authors state that the objective is to "determine the most effective physical activity intervention." I believe more general language about reviewing available studies in this arena would be more fitting as an objective. • The presentation of hypotheses leads the reader to expect statistical tests of these hypotheses. Since this is not what this study accomplished, I think that framing the hypotheses as areas of exploration may be more fitting. • Pg 6: Line 15-16: Please indicate what type of effect on weight loss (e.g., positive, negative) these rates are associated with. • Please provide citations for the feasibility work in progress by the study team if possible. • The authors conclude with and emphasize as key take homes that research is needed on the timing, methods, and content of interventions as well as more engagement with stakeholders. Both of these pghs were a surprise as a reader. If these were extracted in the review, this should be included in the introduction and methods. If they were not, then these conclusions do not follow logically. • There are several punctuation and spacing errors throughout the manuscript. There are also a few grammatical errors such as run-on sentences. I noted the following:

	• Consider revising “adherence” focus to fidelity and adding a focus on feasibility since the manuscript reports on compliance rates (adherence/fidelity) and retention rates (feasibility). Furthermore, recruitment (a fidelity measure) is mentioned in the discussion. • Page 2: Make the “1” after “McGregor” superscript. • Page 2: Add telephone number to corresponding author information, or remove “Tel.” • Abstract: Please use periods consistently at the end of each section. • Abstract: Move the setting info (“Systematic review of PA intervention RCTs in postnatal women”) up and integrate into objectives. • Abstract: Please remove “community and secondary care settings” from the participant info and place in the setting section. • Pg 3, Line 17: Rephrase “PA interventions used were highly varied, targets…” to either o “PA intervention components were highly variable with targets…” o “Types of PA interventions used varied highly with targets…” • Pg 3, Line 17: Please add “minutes” between “PA” and “per week” • Pg 3, Line 24: Remove “interventions” and change “on” to “as” • Pg 3, Lines 28-29: Revise sentence to remove the comma after “further” and state “Further high-quality research that reports adherence and LTFU in needed in order to better understand how and when to deliver postnatal PA interventions to benefit physical and mental health.” • Pg 3, Lines 48-49: Revise sentence in similar manner as above. • Pg 3, Line 50: Add a period after “children” • Pg 3, Line 52: Please use full term for PPI instead of the acronym since this is the first time PPI is used. Also add a period after “studies” • Pg 3, Line 54: Add a period after “professionals” • Pg 4, Lines 9-10: Change font size and/or type of everything after “week;” to match the rest of the document. • Pg 4, Line 18: Remove space between “intensity” and comma. • Pg 4, Line 31: Change “Officers recommends” to either “Officers recommend” or “Officer recommends” • Pg 4, Lines 55-59: Please indicate the year(s) during which the search was conducted. • Pg 4, Line 57: Change to “files 3, 4 and 5) without restrictions on publication date. …” • Pg 5, Line 3: Please indicate if title/abstract review was done in duplicate (as opposed to SS and EM each screening half). • Pg 5, Line 19: Please indicate if data extraction was done in duplicate (as opposed to SS and EM each extracting data from half of included studies). If this was a duplicate process, please indicate how discrepancies were resolved. • Pg 5, Line 24-30: Please indicate if independent risk of bias and GRADE assessments were compared to one another. If yes, please describe how discrepancies were resolved. If no, please specify how ratings were applied (e.g., ratings were averaged, only ratings from SS were used, etc.). • Pg 5, Lines 25-26: Please indicate if the ROB 1, ROB 2, or ROBINS-I Cochrane tool was used. • Pg 5, Lines 27-28: Please state if this is only a portion or all domains from the Cochrane Collaboration’s tool for assessing risk of bias. If only a portion, please indicate why certain domains were excluded.

	• Pg 5, Line 29: the citation for the GRADE criteria (#13) is a reference for the MOOSE guidelines, which do not mention GRADE methods. Please use a more appropriate citation. • Pg 5, Line 37: Add space between “Figure” and “1”; Remove space after close parentheses. • Pg 5, Line 44: Use PA abbreviation or a lowercase “a” for “activity” instead of “Physical Activity…” • Pg 5, Line 48: Add “minutes” between “PA” and “per week” • Pg 5, Line 49: Remove “per week” • Pg 6, Line 57: Add a space before “and” • Pg 6, Line 3 and 4: Add a space after closed parenthesis. • Pg 6, Line 15: Add space between “intervention(16,34).” • Pg 6, Line 16: Add “an” between “showed” and “effect” • Pg 6, Line 18: Chance comma to period after “(18,21,25)” and capitalize “t” in “the” • Pg 6 Line 20: Change comma after “both” to a semicolon and capitalize the “a” in “adherence” • Pg 6, Line 26: Add a space between clothes parenthesis and “appear” • Pg 6, Line 27: Add a space between clothes parenthesis and “and” • Pg 7, Line 11-12: Change “physical activity is unlikely to impact on breastfeeding…” to either “physical activity is unlikely to have an impact on breastfeeding…” or “physical activity is unlikely to impact breastfeeding …” • Pg 7, Line 26: Change “into” to “on” • Pg 7, Line 27: Remove “recruitment” or add recruitment as a variable of interest throughout the rest of the manuscript. • Pg 7, Lines 28-30: Reword to “Weight gain is encouraged in pregnancy, but complications such as excessive gestational weight gain, preeclampsia, or gestational diabetes may help identify women likely to benefit from PA interventions to improve their long-term health.” • Pg 7, Line 33-36: Reword to “Public health approaches which encourage and enable women to increase their PA after having a baby could have broader population impacts with benefits seen in short- and long-term risk of chronic diseases; However, studies to date have not consistently demonstrated these effects.” • Pg 7, Line 37: Add “the” between “in” and “design” • Table 1: Please add EPDS to abbreviations key.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Reviewer Name: Bee Tan
Institution and Country: University of Leicester, England, UK.
Please state any competing interests or state ‘None declared’: None declared.

This systematic review addresses a very important issue. The review is well performed, employing appropriate methodology.

Reviewer: 2
Reviewer Name: Daria Kołomańska-Bogucka
Institution and Country: Jagiellonian University Collegium Medicum, Poland
Please state any competing interests or state ‘None declared’: None declared

The Authors present a systematic review of 24 studies about physical activity of women during postpartum period. This is a very important topic, because properly selected physical activity has a positive effect on the course of pregnancy, delivery and postpartum period.
The authors have pre-registered the review on PROSPERO. They also presented a detailed way of browsing individual databases. Information on the individual phases of the study is presented in the PRISMA diagram. The studies qualified for the review are presented in detail in the tables.
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5) In the paragraph Search strategy and selection criteria you wrote that "…publication date was not restricted", but could you add information in what period (month, year) the review was performed?

Thank you, these changes are tracked in the revised manuscript.

Reviewer: 3
Reviewer Name: Taren Swindle
Institution and Country: University of Arkansas for Medical Sciences, United States
Please state any competing interests or state 'None declared': Mentored review in collaboration with Audrey Martinez

Thank you for the opportunity to review "Postnatal Exercise Interventions: A Systematic Review of Adherence and Effect." This is a systematic review focusing on 24 RCTs of physical activity interventions during the first-year post-partum. The primary outcomes are adherence (operationalized as compliance and retention rates) and effectiveness. This systematic review showed inconsistent results for most outcomes, besides adverse events and negative impacts on breastfeeding or infant growth. There was a lack of reporting on adherence measures and conclusions are generally based on low quality studies at a high risk of bias. Appropriate detail is given for the methods of the review, and justification for exploration of this topic is clear and fitting. I have the following comments, questions, and suggestions to improve the paper:
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Thank you, search updated.

• No meta-analysis was performed, and this systematic review focuses on intervention studies, yet they used the Meta-Analysis of Observational Studies in Epidemiology (MOOSE) guidelines to report findings. I believe PRISMA would be more appropriate. I recommend requesting a revised manuscript that uses PRISMA instead of MOOSE guidelines.

Thank you, included.

• The authors state that sub-group analyses were conducted. It is not clear what is meant by analyses in this case. I wonder if instead it was an examination of patterns.

Comparison of supervised/unsupervised studies.

• The authors state that "supervised studies appeared less effective than unsupervised studies." For this comparison and others in the studies, could the authors be clear about the thresholds or specific data used to draw these conclusions?

Amended: 'Supervised exercise interventions did not appear to be more likely to show effect on the various outcomes tested than unsupervised interventions. Studies evaluating unsupervised interventions had lower LTFU and higher adherence, which increases our confidence in the outcomes of the studies. This was unexpected and may reflect that women may feel more empowered when allowed to exercise on their own terms; self-efficacy may lead to greater efficacy of the intervention.'

• The take home messages do not include any findings from the study.

Not included now
• The authors state that the objective is to "determine the most effective physical activity intervention." I believe more general language about reviewing available studies in this arena would be more fitting as an objective.

Amended, 'To review available studies on postnatal physical activity interventions, we conducted a systematic review to determine adherence to postnatal physical activity interventions, and the effect of the interventions on the outcomes investigated in the studies identified. The results would also inform healthcare professionals who advise women on the most efficacious PA interventions for desired effect.'
• The presentation of hypotheses leads the reader to expect statistical tests of these hypotheses. Since this is not what this study accomplished, I think that framing the hypotheses as areas of exploration may be more fitting.

Amended, 'We wished to explore whether adherence to PA interventions would be greater for supervised (in person supervision or with monitoring of PA e.g. a pedometer) vs. unsupervised interventions; and that the effects of PA interventions on the primary outcomes of studies would be related to adherence.'
• Pg 6: Line 15-16: Please indicate what type of effect on weight loss (e.g., positive, negative) these rates are associated with.
• Please provide citations for the feasibility work in progress by the study team if possible.
• The authors conclude with and emphasize as key take homes that research is needed on the timing, methods, and content of interventions as well as more engagement with stakeholders. Both of these pghs were a surprise as a reader. If these were extracted in the review, this should be included in the introduction and methods. If they were not, then these conclusions do not follow logically.

Amended

In response to the multiple suggestions below, we have reviewed the manuscript and made amendments where appropriate.
• There are several punctuation and spacing errors throughout the manuscript. There are also a few grammatical errors such as run-on sentences. I noted the following:
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